Parth Shah

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EDUCATION

UNIVERSITY OF CALIFORNIA, DAVIS
BS in Computer Science and Engineering
Class of '23 | Davis, CA
GPA: 3.7 / 4.0

SKILLS

Proficient with:
• Python • Java • C++ • TypeScript • LangGraph • Gemini & Ollama • MCP • FastAPI • ReactJs • NodeJs • Git & Docker • AWS • Bash • LaTeX • SQL

AWARDS

2026 2nd AgentX - AgentBeats
2024 Bias for Action - Amazon
2023 Best Use of GitHub - HackDavis
2022 Scholars Award - Chevron
2022 1st - HackDavis
2022 Best Local Hack - SacHacks
2021 2nd - HackDavis
2020 Top 30 - PennApps XXI
2020 Top 20 - Google Solution
2020 Best Social Good - HackDavis

EXPERIENCE

AMAZON | Software Development Engineer 2
July 2023 - Present | Sunnyvale, CA

- Own a cross-org initiative automating operational ticket investigation with AI agents, built on three layers: an operational memory that clusters past tickets by symptom and root cause to generate agentic skills from SOPs, a semantic code graph that makes codebases agent-friendly, and a knowledge layer of wikis, code reviews, and design docs.
- Validated generated skills against a holdout set with an actor–critic loop, reaching 90% accuracy on holdout tickets; skills are discovered dynamically at runtime across teams. Live for 3 teams (30+ engineers) and expanding to multiple orgs, cutting root-cause time from 30–60 minutes to ∼10 minutes across 1,200+ tickets a year.
- Spearheaded the infrastructure for World Catalog, Amazon's pioneering agent-first global catalog. Engineered data systems to ingest and harmonize products beyond the Amazon ecosystem, powering next-generation AI search tools.
- Architected the Progressive Listing Experience (PLE), integrating LLMs to autonomously scrape and map external seller URLs to the complex Amazon catalog schema, drastically reducing seller onboarding friction.
- Led a ground-up architecture rewrite of the catalog contribution ingestion pipeline to support modern asynchronous processing required for advanced LLM workflows enhancing end-to-end system traceability and security, while completely overhauling the CI/CD deployment infrastructure to reduce system build times by over 80%.
- Designed and launched 3 net-new microservices utilizing a highly extensible plugin architecture for catalog data validation, creating an execution engine with over 200 custom rules for marketplace translation and image enrichment.

AMAZON | SDE Intern, Item Pipeline
June 2022 - Sep 2022 | Sunnyvale, CA

- Engineered a custom Code Analyzer to automate bug detection and enforce strict coding standards, directly improving organizational code quality and reducing operational burden.
- Successfully championed the tool's integration into the CI/CD pipelines of 8 distinct teams, achieving widespread adoption by over 100 active engineers.
- Architected and deployed scalable, secure serverless microservices for the Item Pipeline utilizing Java, TypeScript, AWS Lambda, Amazon SNS, and VPCs.

PROJECTS

NAAMSE | GitHub | Paper
Jan 2026 - Present

- Engineered the scalable core infrastructure for an evolutionary AI red-teaming framework and fuzzer, optimizing high-throughput execution pipelines for adversarial prompt injection.
- Architected the underlying data systems to support real-time attack mutation using genetic algorithms, enabling continuous balancing of Attack Success Rate (ASR) and False Positive Rate (FPR).

- Won 2nd place out of 1,500+ submissions at the Berkeley RDI AgentX competition; framework infrastructure was subsequently forked by Mozilla's 0din team.

HASHIRU | GitHub | Paper
Early 2026

- Architected a budget-aware multi-agent orchestration framework (Hierarchical Agent System for Hybrid Intelligent Resource Utilization) enabling highly distributed workload management.
- Built robust cost-optimization controls to strictly manage resource compute constraints, inference expenses, and token budgeting across parallel LLM agent execution streams.

GEM5 VISION | Website | GitHub | Poster
Jan 2023 - June 2023

- Engineered advanced search functionality and a comprehensive categorization system for the gem5 Resource Infrastructure, enhancing resource discovery for computer architecture researchers.
- Architected extensible database integrations to support diverse storage and retrieval options, providing developers with robust tools for managing simulator resources.

STADIA WIRELESS | GitHub
July 2022

- Reverse-engineered and bypassed Google Stadia Controller hardware constraints, implementing a websocket-based architecture to enable universal wireless connectivity.
- Open-sourced the solution, accumulating 450+ GitHub stars and earning features in major tech publications including The Verge and Ars Technica.